

Supply Chain Management

What is Supply Chain Management (SCM)?

Supply Chain Management (SCM) is the process of **planning, managing, and optimizing the flow of goods, information, and money**—from raw materials all the way to the final customer.

👉 A typical flow looks like:

Supplier → Manufacturer → Warehouse → Distributor → Retailer → Customer

⚙️ Key Components of SCM

1) 📦 Procurement (Sourcing)

- Buying raw materials from suppliers
- Choosing vendors, negotiating prices

2) 🏭 Production (Manufacturing)

- Converting raw materials into finished goods
- Planning capacity, schedules, quality

3) 📊 Inventory Management

- Controlling stock levels across warehouses
- Avoiding overstock and stockouts

4) 🚚 Logistics & Distribution

- Transportation, warehousing, delivery routes
- Getting products to markets/customers

5) 🔄 Demand Planning & Forecasting

- Predicting future demand using data
- Aligning supply with expected sales

Why SCM Matters

- 💰 **Cost reduction** (less waste, better purchasing)
- 🚀 **Faster delivery** (optimized routes & stock placement)
- 😊 **Customer satisfaction** (products available on time)
- 📈 **Better decisions** (data-driven planning)

🧠 STEP 1: Demand Planning

👉 Question:

- How much product will customers need?

What happens:

- Analyze past sales
- Forecast demand (monthly/quarterly)

📊 **Your Dashboard:**

- Sales trend (line chart)
 - Seasonal demand pattern
-

🛒 STEP 2: Procurement (Buying from Supplier)

👉 Question:

- When should we order stock?

What happens:

- Supplier selection
- Purchase orders created

📊 **Your Dashboard:**

- Supplier performance

- Lead time analysis
-

STEP 3: Inbound Logistics (Supplier → Warehouse)

 Question:

- How goods arrive?

What happens:

- Transportation arranged
- Goods received at warehouse

 **Your Dashboard:**

- Delivery delays
 - Supplier reliability
-

STEP 4: Warehouse Operations

 This is VERY IMPORTANT 🔥

Process inside warehouse:

Receiving → Putaway → Storage → Picking → Packing → Shipping

A. Receiving

- Goods arrive
 - Checked for quality & quantity
-

B. Putaway

- Items stored in correct location
-

C. Storage

- Organized based on demand
 - Fast-moving items → easy access
-

D. Picking

- Items picked when order comes
-

E. Packing

- Packed for delivery
-

F. Shipping

- Sent to customer
-

Your Dashboard:

- Stock levels
 - Warehouse efficiency
 - Order processing time
-

STEP 5: Inventory Management

Core logic:

- How much stock do we have?
 - When to reorder?
-

Key concepts:

- Safety Stock
- Reorder Point

- Inventory Turnover
-

Your Dashboard:

- Low stock alerts 
 - Inventory turnover 
 - Overstock detection 
 -  **Inventory turnover Simple Meaning**
 - It shows **how fast your stock is moving**
 - Helps you understand **demand and efficiency**
 - Used a lot in **inventory, warehouse, and retail analytics**
-

-  **Example**
 - Cost of Goods Sold (COGS) = 100,000
 - Average Inventory = 20,000
 -  Inventory Turnover = 5
 -  This means your inventory is sold and restocked **5 times in that period**
-

STEP 6: Outbound Logistics (Delivery)

Question:

- How products reach customer?

What happens:

- Route planning
 - Delivery execution
-

Your Dashboard:

- Delivery performance
 - Order fulfillment rate
-

STEP 7: Customer Delivery & Feedback

 Final step:

- Customer receives product
-

Metrics:

- On-time delivery
- Customer satisfaction

Key Differences (Very Important)

Feature	Warehouse 	Inventory 
Meaning	Storage place	Goods/products
Type	Physical infrastructure	Items/data
Focus	Space & operations	Quantity & value
Example	Building in Kathmandu	500 units of laptops

Imagine a Daraz warehouse:

- The **building** = Warehouse 
- The **products inside (phones, clothes)** = Inventory 

 Without warehouse → nowhere to store

 Without inventory → nothing to manage

Core Domain Areas You MUST Prepare

1 Inventory Fundamentals

What is inventory and why is it important?

What interviewer expects:

- Inventory = goods stored for future sale/use
 - Balance between:
 - Stock availability (avoid stockout)
 - Holding cost (avoid overstock)
-

What is Safety Stock?

Expected Answer:

- Extra stock kept to handle demand uncertainty or delays

 Formula-level understanding helps:

- Depends on demand variability + lead time
-

What is Reorder Point (ROP)?

Key concept:

When to reorder stock before it runs out

 Formula:

ROP = (Average Daily Demand × Lead Time) + Safety Stock

💡 Link to your project:

You already used this in **low stock detection**

2 Demand vs Inventory Thinking

? How do you identify understocked products?

Strong Answer:

- Compare:
 - Sales velocity (demand)
 - Current stock level

👉 Example logic:

- High demand + low stock = understock risk
-

? How do you identify overstocked items?

Strong Answer:

- Low sales but high inventory
- Slow-moving or dead stock

💡 Business angle:

- Leads to holding cost, warehouse blockage
-

3 Key KPIs (VERY IMPORTANT)

? What is Inventory Turnover?

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$
$$\text{Inventory Turnover} = \frac{\text{Average Inventory}}{\text{Cost of Goods Sold}}$$

Explain like this:

- Measures how fast inventory is sold

- High turnover = efficient
 - Low turnover = slow-moving stock
-

? What is Days Inventory Outstanding (DIO)?

👉 Formula:

$$\text{DIO} = (\text{Average Inventory} / \text{COGS}) \times 365$$

Meaning:

- Number of days inventory sits before being sold
-

? What is Stockout?

Answer:

- When inventory = 0 but demand exists
👉 Leads to lost sales
-

? What is Carrying Cost?

Answer:

- Cost of holding inventory:
 - Storage
 - Insurance
 - Damage risk
 - Opportunity cost
-

4 Business Scenario Questions (VERY COMMON)

? A product has high sales but low stock. What will you do?

Expected Thinking:

- Increase reorder frequency

- Adjust safety stock
 - Analyze supplier lead time
-

? A product has high stock but low sales. What will you do?

Strong Answer:

- Run promotions/discounts
 - Bundle products
 - Reduce future procurement
-

? Sales suddenly drop. How do you analyze?

Approach:

- Check seasonality
 - Check stock availability
 - Check pricing changes
 - External factors (market trends)
-

5 SQL + Business Combined Questions

? How would you find top-selling products?

👉 Expectation:

```
SELECT product_id, SUM(quantity_sold)
FROM sales_data
GROUP BY product_id
ORDER BY 2 DESC;
```

? How to detect low stock products?

👉 Logic:

WHERE stock_level < reorder_point

? How to compare demand vs stock?

👉 Join + aggregate:

- SUM(quantity_sold) vs stock_level
-

6 Supply Chain Concepts (Moderate Level)

? What is Lead Time?

- Time between ordering and receiving stock
-

? What is EOQ (Economic Order Quantity)?

👉 Basic understanding is enough:

- Optimal order quantity to minimize total cost.
 - **EOQ (Economic Order Quantity)** is the **optimal quantity of items a business should order each time** to minimize total inventory costs.
-

? What is Bullwhip Effect?

Strong answer:

- Small demand changes → large supply chain fluctuations
-

7 Power BI + Business Questions

? What dashboards would you build for inventory?

Answer:

- Stock status (low / ok / overstock)
- Sales vs inventory trend

- Top products
 - Inventory turnover KPI
-

What filters are important?

- Time (date)
 - Product category
 - Warehouse
-

How YOU should answer (Important Tip)

Don't say only definition. Always structure:

-  Definition
 -  Formula (if applicable)
 -  Business impact
 -  Example
-

Example (Perfect Answer Style)

Q: What is Inventory Turnover?

Inventory turnover measures how efficiently inventory is sold over a period.

It is calculated as cost of goods sold divided by average inventory.

A high turnover indicates strong sales and efficient inventory management, while a low turnover may indicate overstocking or weak demand.